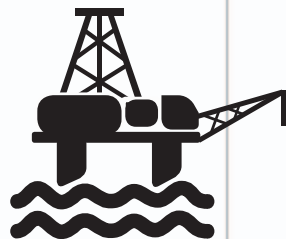




STRUCTURAL ENGINEER

JOB DESCRIPTION

Structural engineers help to design buildings and infrastructure, such as bridges, railroads, dams, and tunnels. They analyze the forces that a structure may face, such as winds, people loads, and traffic, and work with architects and civil engineers to ensure that it is built to required standards of strength and safety.



SALARY

Entry level structural engineer ★★☆☆☆

Senior structural engineer ★★★★★

INDUSTRY PROFILE

Growing industry with opportunities across the globe • Employers range from governments to a variety of contractors and consultancies

AT A GLANCE



YOUR INTERESTS Engineering • Mathematics • Physics • Information Technology (IT) • Design • Geography • Drawing and modelmaking



ENTRY QUALIFICATIONS A degree in structural or civil engineering from an accredited program is required. Advanced degrees may be needed.



LIFESTYLE Regular office hours are the norm, although engineers may need to be on call to deal with emergencies, such as damaged or unstable buildings.



LOCATION Most engineers divide their time between an office and construction sites. They may occasionally need to travel around the country for work.



THE REALITIES Construction is one of the first sectors to be affected in an economic slump. Sites are often dusty and noisy, and can be dangerous.

RELATED CAREERS

- ▶ **CIVIL ENGINEER** *see pp. 176–177*
- ▶ **ARCHITECT** *see pp. 194–195*
- ▶ **BUILDING CONTROL OFFICER** Ensures that building regulations and other laws are followed in the design and construction of houses, offices, and other buildings. Building control officers also make sure that property alterations, such as extensions and conversions, meet all the current regulations.
- ▶ **COMPUTER-AIDED DESIGN (CAD) TECHNICIAN** Uses computer design software to create plans for buildings and machinery. CAD technicians can work in a range of industries, including construction, manufacturing, and engineering.

Demand for structural engineers is increasing, partly due to growing numbers of aging buildings.